Module 6 Assignment



This picture shows that the source is real.

1. Portability. Why does your European-derived score lose accuracy in other ancestries? Point to LD and allele-frequency differences.

This analysis shows that there are differences in the predictive performances of PBS based on European-derived scores across different ancestry groups. From the 3.4 in the notebook, the EUR group shows the highest AUC is 0.672, while the AFR group is 0.493, which is lowest and close to the random discrimination, which is 0.5**.** The AUC of EAS, SAS, and AMR are separately 0.575, 0.611 and 0.555. These results mean the portability of PRS is limited across the ancestry groups.

The one reason is the difference in linkage disequilibrium (LD) patterns in different ancestry groups. Some SNPs in PRS are markers for the nearby associated disease variants. In the European ancestry population, these marker SNPs may have strong correlations with associated disease variants, but it may be weak in other ancestry groups. Therefore, the PRS in these populations may not capture enoughly effective genetic risk information.

The difference in allele frequency may also cause the reduction of PRS portability. The appearing frequency of a specific allele may be different in different ancestry groups, which will influence the distribution of genetic variants and the predictive information captured by

PRS. So, PRS developed based on European ancestry data may not show the same prediction effect in other populations.

However, since 1000 Genomes does not provide disease diagnosis labels, the phenotype used in this study is simulated. Therefore, these AUC results only have teaching significance and cannot directly represent the real clinical predictive performance, nor can they prove that there are inherent biological differences in diseases between different ancestry groups.

2. One threshold, many groups. Using your Section 3.3 numbers, is a single top-10% cutoff fair? What happens if a clinic applies it unchanged?

From 3.3 in the notebook, we set a threshold of the first 10% according to the PRS distribution of all research subjects. However, after using the same threshold, the proportion of people classified as "high risk" in different ancestral groups varies greatly. About 28.0% of the AFR group above the cutoff, while 2.2% of EAS above the cutoff. Also, there are 7.8% of EUR, 4.3% of SAS, and 6.3% of AMR above the cutoff.

A single top-10% cutoff has fairness concerns that we cannot judge whether it is fair just depends on these figures. The reason is that we just know how many people are above the cutoff rather than knowing whether these people have disease and will have disease or not in the future. If a clinic applies this cutoff, there will be more AFR group people marked as high risk and then accept more screening, while the EAS group people marked less and there are some people who need to be concerned cannot be screened out.

3. Discrimination vs calibration. What did each tell you (3.4, 3.5), and where did they disagree?

From the 3.4 in the notebook, we use the AUC to evaluate the discrimination of PRS which is whether PRS can distinguish between simulated cases and controls. The EUR group's AUC is 0.672 that means PRS has a certain discrimination. The AUC of AFR is 0.493 close to the random discrimination. This means PRS has better ability to distinguish in EUR groups than AFR.

From the 3.5 in the notebook, we use Calibration to compare the disease probability predicted by the model and the ratio of simulated cases actually observed. It fits the risk model in EUR only, applies it to all ancestry groups, and draws one calibration curve per ancestry. The plot shows deviation between predicted probabilities and observed case proportions, especially in AFR.

Discrimination and Calibration measure two different aspects of the prediction model, which is more obvious in the EUR group. Although the AUC of EUR is 0.672 that means the model has certain discrimination, there is still a deviation between the predicted probability and the observed proportion of simulated cases in the 3.5 notebook part.

4. Clinical utility and limits. Where could the score help, where should it not be used, and what would you need before deploying it? Note that the phenotype here is simulated, and name the real data you would need to validate for real.

The PRS can be a useful additional tool for identifying groups that may require further risk assessment or preventive attention. From 3.6 in the notebook, the AUC age only shows 0.562 and it improves to 0.618 after adding the PRS, which means the discrimination improves with PRS.

However, the PRS should not be used alone for disease diagnosis, nor should it be directly used as the only evidence for making clinical decisions. The notebook 3.4 and 3.5 shows

differences in discrimination and calibration in ancestry groups, which means PRS may not provide reliable predictions for each group. So , we cannot judge whether a person is sick or needs treatment based on his PRS score alone.

Before deploying the score, we need to use the real data of patients to validate the model, including real disease diagnoses, genetic information, patient information and so on. We also need to test the discrimination and calibration of the model in other populations.

5. Artifact versus biology. The score runs near chance in the African-ancestry group here, partly because the simulation's LD proxy for APOE ε4 is almost absent in that group. Explain why that is a property of this simulated dataset rather than evidence that ε4 is unimportant in African-ancestry people.

From the notebook 3.4, the AUC of AFR group id 0.493, which is close to the random discrimination. The explanation for this is that the simulation's LD proxy for APOE ε4 is almost absent in that group, making it hard to capture genetic information. However, this result shows limitations of the simulated dataset, not the biological importance of APOE ε4 itself. A weak proxy doesn’t mean APOE ε4 has no influence in disease risk in the African-ancestry group. So, the low AUC cannot be the evidence proving APOE ε4 is not important in that group.